

QUESTION BANK FOR IV SEMESTER

Subject Code: CSL46

TERM: Feb – May 2026

I.A. Marks: 50

Subject Name: Design and Analysis of Algorithms Laboratory

Exam Hours: 03

Credits: 0:0:1

Exam Marks: 50

Sl No.	List of Programs														
1.	In a university examination cell, student marks are stored in a database in random order. The examination officer wants to sort the marks efficiently by repeatedly dividing the list into smaller sublists, sorting them, and then combining the results to obtain a fully sorted list. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to sort the given list of marks[78, 91, 65, 83, 70, 95, 69, 88] and display the sorted output with the space and time complexity.														
2.	Same as first. Little advanced. A university examination system maintains two separate arrays: - One array stores the CGPA values of students, which are unsorted - Another logical grading rule is used to assign letter grades based on CGPA as follows: <table border="1"> <thead> <tr> <th>CGPA Range</th> <th>Grade</th> </tr> </thead> <tbody> <tr> <td>CGPA $\geq$ 9.0</td> <td>O</td> </tr> <tr> <td>CGPA $\geq$ 8.0 and $<$ 9.0</td> <td>S</td> </tr> <tr> <td>CGPA $\geq$ 7.0 and $<$ 8.0</td> <td>A</td> </tr> <tr> <td>CGPA $\geq$ 6.0 and $<$ 7.0</td> <td>B</td> </tr> <tr> <td>CGPA $\geq$ 5.0 and $<$ 6.0</td> <td>C</td> </tr> <tr> <td>CGPA $<$ 5.0</td> <td>F</td> </tr> </tbody> </table> The examination cell wants to merge the CGPA values ([7.8, 9.2, 6.4, 8.1, 4.9, 5.6, 7.0, 8.9]) with their corresponding grades and store them in a single array structure, such that each entry contains CGPA and its grade together, arranged in descending order of CGPA. Merged and sorted result: [9.2 - O, 8.9 - S, 8.1 - S, 7.8 - A, 7.0 - A, 6.4 - B, 5.6 - C, 4.9 - F]	CGPA Range	Grade	CGPA $\geq$ 9.0	O	CGPA $\geq$ 8.0 and $<$ 9.0	S	CGPA $\geq$ 7.0 and $<$ 8.0	A	CGPA $\geq$ 6.0 and $<$ 7.0	B	CGPA $\geq$ 5.0 and $<$ 6.0	C	CGPA $<$ 5.0	F
CGPA Range	Grade														
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3.	An online shopping platform stores product prices in an unsorted manner. The system selects a reference price and rearranges the prices such that all prices lower than the reference appear on one side and higher prices on the other side, repeating this process until the entire list is sorted. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to sort the product prices and display the execution time.														
4.	Given a set of men and women design and implement Gale–Shapley algorithm to determine the stable set of marriages among them. Assumptions: Men propose first according to their preference list. Women can choose a better partner based on the preference. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java														

program to match the preferences perfect and stable.

Men's preference list

	1	2	3	4
A	W	X	Y	Z
B	X	W	Y	Z
C	Z	W	Y	X
D	Z	Y	X	W

Women's preference list

	1	2	3	4
W	A	B	D	C
X	B	C	A	D
Y	C	D	A	B
Z	C	D	B	A

A software development project consists of multiple tasks, where certain tasks must be completed before others can begin. These task dependencies are represented using a **directed graph**, where each node represents a task and each directed edge represents a dependency between tasks. To ensure correct execution of the project, the project manager wants to determine a **valid order of task execution** such that no task is started before all the tasks on which it depends are completed. The solution should explore tasks deeply along dependency paths and return back when no further dependent tasks are available, continuing this process until all tasks are scheduled.

Number of tasks = 6

Dependencies:

$5 \rightarrow 2$

$5 \rightarrow 0$

$4 \rightarrow 0$

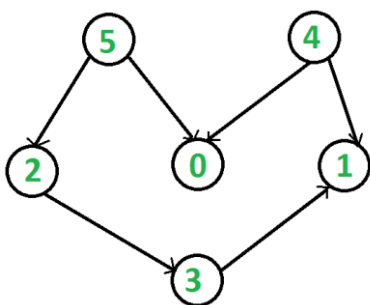
$4 \rightarrow 1$

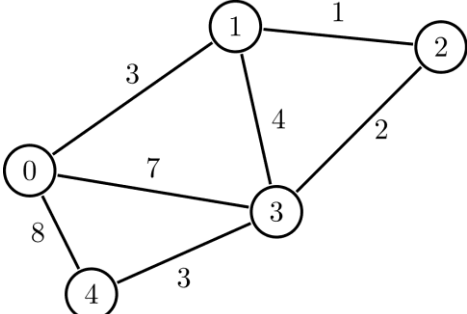
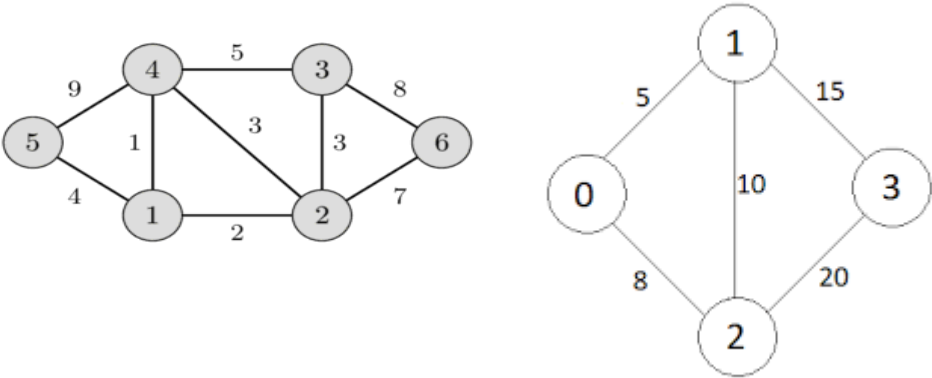
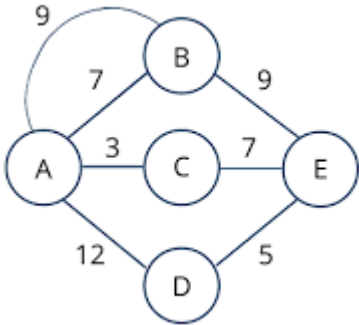
$2 \rightarrow 3$

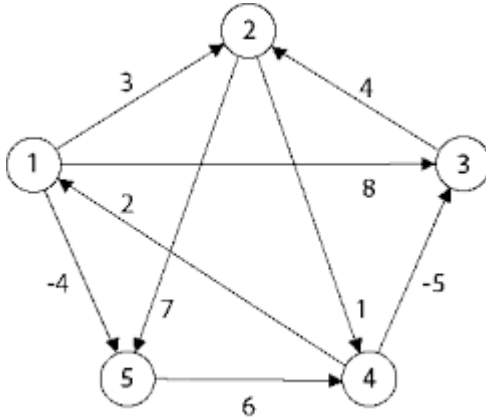
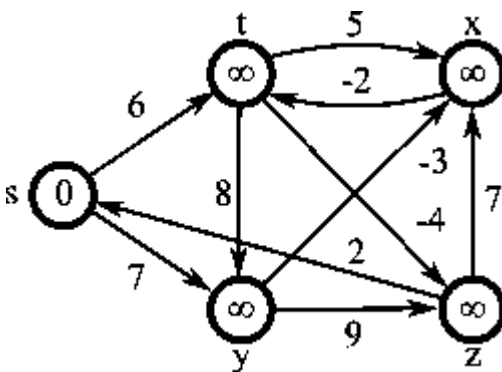
$3 \rightarrow 1$

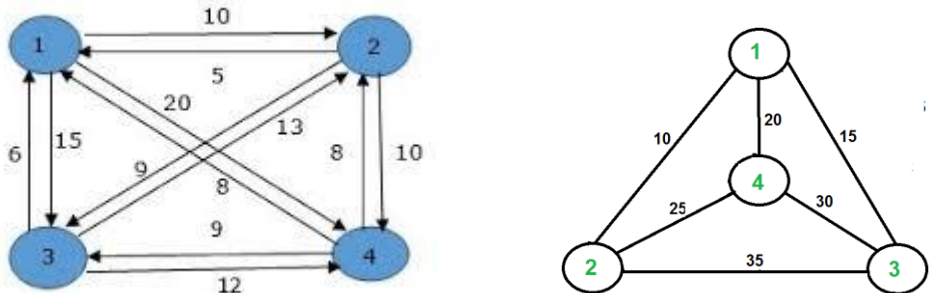
5.

Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to determine a valid task execution order. Also, analyze the time and space complexity. **Or graph to be kept**



6.	A truck driver is given a set of locations with distances between them. The company strictly orders that the truck must start from a specific source location and reach all other locations using the minimum possible distance. . Identify the technique and algorithm to solve the given problem. Implement the algorithm using java find the shortest distance from the source (Source = 0) to all other locations. Also, evaluate time and space complexity. 
7.	A city planning department wants to connect all major junctions using roads such that the total construction cost is minimum. The construction can start from any junction. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to form a minimum-cost road network. Also,analyze time and space complexity. 
8.	A telecom company plans to lay optical fiber cables between cities. Each possible connection has a different cost. The goal is to connect all cities with minimum total cost without forming cycles. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to generate a minimum-cost communication network. 
9.	A system administrator needs to search for a specific error keyword in a large system log file(Text = "DATA STRUCTURES AND ALGORITHMS").To reduce unnecessary character comparisons, the algorithm should utilize information derived from the pattern(Pattern = "ALGO"). Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to efficient perform string matching. Also, analyze time and space complexity.

10.	A plagiarism detection system searches (Pattern = "ENGINE") for a particular word in large documents(Text = "COMPUTER SCIENCE AND ENGINEERING").The system compares characters from right to left and skips multiple characters whenever a mismatch occurs. "). Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to efficient perform string matching Also, evaluate time and space complexity.															
11.	A network administrator wants to compute the shortest communication path between every pair of servers in a network. "). Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to compute all-pairs shortest paths. Also, analyze time and space complexity. 															
12.	A trekking enthusiast carries a backpack with a maximum capacity of 7 kg. Each item has a specific weight and value. The objective is to maximize total value without exceeding the capacity, while avoiding repeated subproblem computation. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to select the most valuable items. Also, evaluate time and space complexity. <table><tr><th>Item</th><th>Weight</th><th>Value</th></tr><tr><td>1</td><td>3</td><td>10</td></tr><tr><td>2</td><td>5</td><td>4</td></tr><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>4</td><td>2</td><td>11</td></tr></table>	Item	Weight	Value	1	3	10	2	5	4	3	6	9	4	2	11
Item	Weight	Value														
1	3	10														
2	5	4														
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13.	A financial transaction network consists of multiple routes, some of which have negative transaction costs. The institution wants to determine the shortest transaction cost from a given source and detect negative-cost cycles. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to compute shortest paths and detect such cycles. Also, analyze time and space complexity. 															

14.	A chess puzzle requires placing queens on an $N \times N$ ($N = 4$) chessboard such that no two queens attack each other. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to display all valid queen arrangements. Also, evaluate time and space complexity .
15.	A financial auditing system checks whether a subset of transaction values adds up exactly to a suspicious target amount. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to determine whether such a subset exists. Also, analyze time and space complexity . Set = {10, 7, 5, 18, 12, 20, 15} Target Sum = 35
16.	A courier company wants to determine the shortest possible delivery route for a delivery agent. The agent must visit each city exactly once and return to the starting city, eliminating routes that exceed the current minimum cost. Identify the technique and algorithm to solve the given problem. Implement the algorithm using java to determine the minimum delivery cost.  The left graph shows a complete graph with 4 nodes (1, 2, 3, 4) and weighted edges. The edges and their weights are: (1,2): 10, (1,3): 6, (1,4): 15, (2,3): 20, (2,4): 13, (3,4): 12. The right graph shows a complete graph with 4 nodes (1, 2, 3, 4) and weighted edges. The edges and their weights are: (1,2): 10, (1,3): 15, (1,4): 20, (2,3): 35, (2,4): 25, (3,4): 30.